Towards Practical and Efficient Federated Learning

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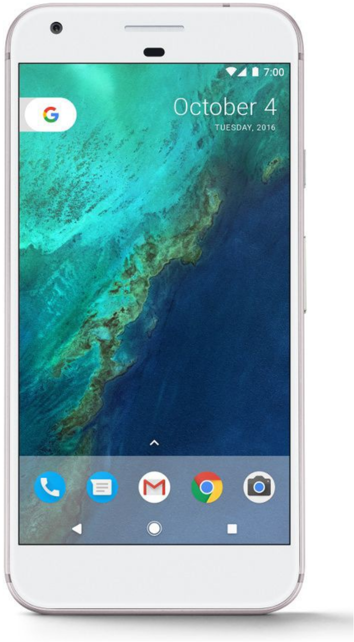
Short Bio

- Interest in distributed systems, machine learning, and networking
- Ph.D. (HKUST) in Comp Sci & Eng (CSE):
  - Improving the performance and efficiency of Distributed/Networked Systems
- Past experience:
  - Senior Researcher (Huawei, HK): Application Driven Networking (ADN)
  - Research Scientist (KAUST, KSA): Distributed ML / Federated Learning

Lead SAYED Systems Lab (https://sayed-sys-lab.github.io)

Federated Learning and ML systems
Distributed and Networked Systems
Computer Networks and Software-Defined Networking
Cloud/Edge Computing and Big Data Analytics

Data lives at the Edge



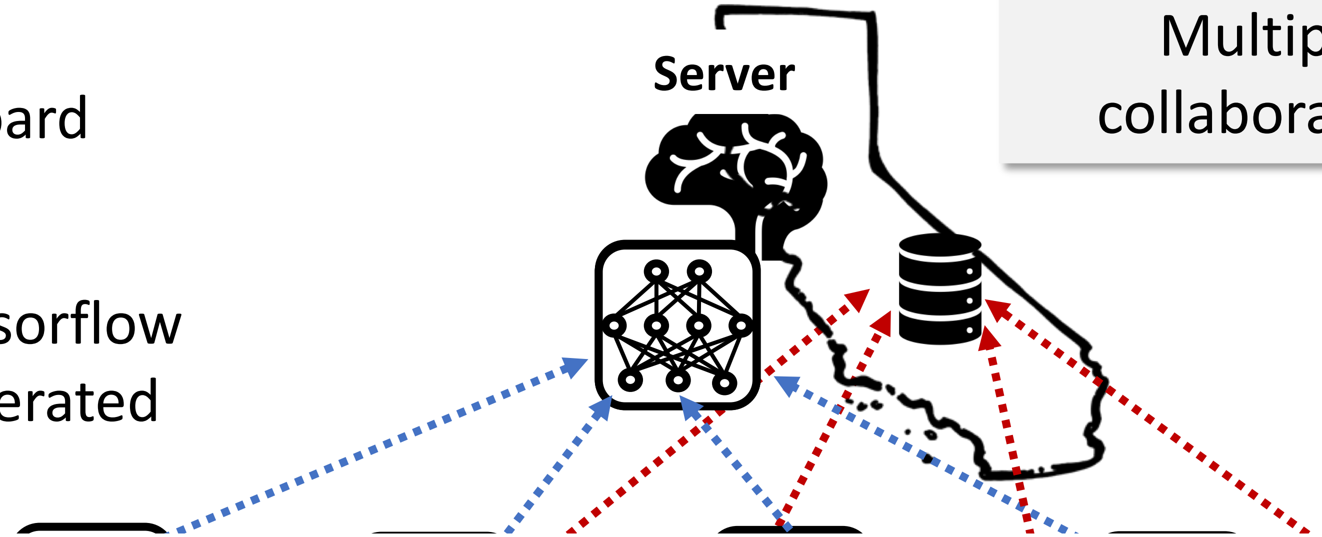
- Billions of phones & IoT devices constantly generate data
- Data enables better products and smarter models
- On-device processing (e.g., inference for mobile keyboards)
  - advanced specialized hardware (e.g., GPU and NPUs on mobile/IoT devices)
- Benefits
  - Improved latency
  - Works offline
  - Better battery life
  - Privacy advantages

What about analytics & learning?

Centralized or Federated Learning

Gboard

Tensorflow Federated

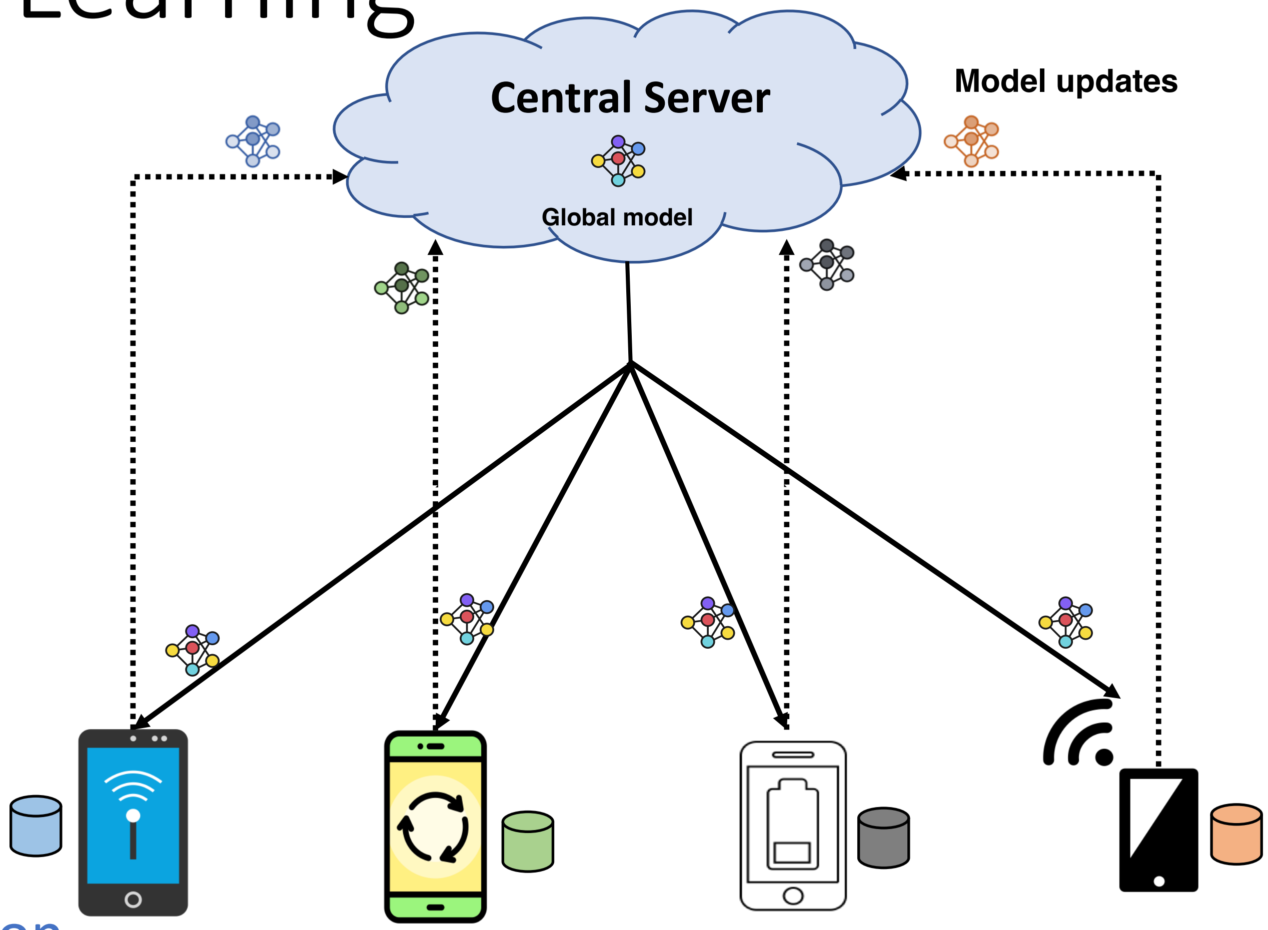


Multiple clients want to collaboratively train a model

Collect DATA vs Collect Models/Gradients?

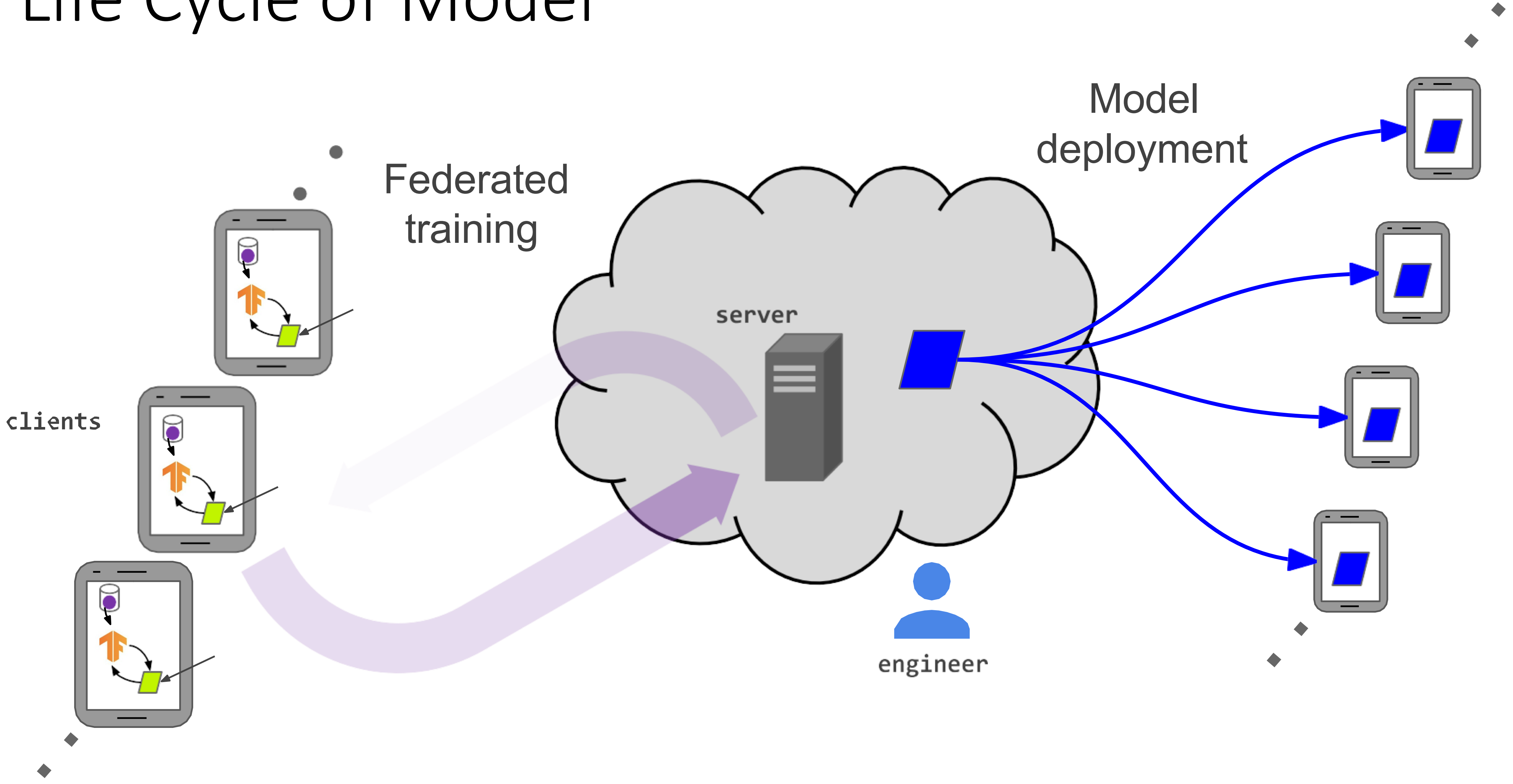
Centralized vs Federated Learning

- Centralized Training:
  - Central (Data) server
  - Expensive data movement
  - Communication-intensive
  - **Privacy concerns**
- Federated Learning:
  - Central (Aggregation) server
  - Model exchange
  - Communication-efficient
  - Differential privacy + secure aggregation



FL is also more favorable for SMEs (min server costs)

Life Cycle of Model



Practical Use-cases of Federated Learning (FL)

What are good applications for FL?

- On-device data is more relevant than server-side data (or lack of it)
- On-device data is privacy sensitive or large to communicate
- Labels can be inferred naturally from user interaction

Gboard: next-word prediction

Using FL, better next-word prediction accuracy: +24%

A. Hard, et al. Federated Learning for Mobile Keyboard Prediction. arXiv:1811.03604

Apple: Voice recognition

MIT Technology Review   Sign in   Subscribe

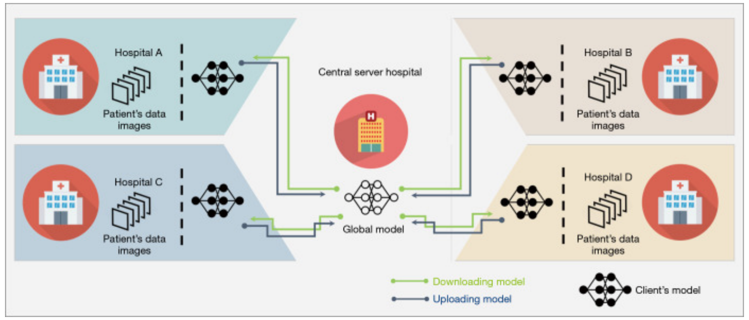
Artificial intelligence / Machine learning

How Apple personalizes Siri without hoovering up your data

The tech giant is using privacy-preserving machine learning to improve its voice assistant while keeping your data on your phone.

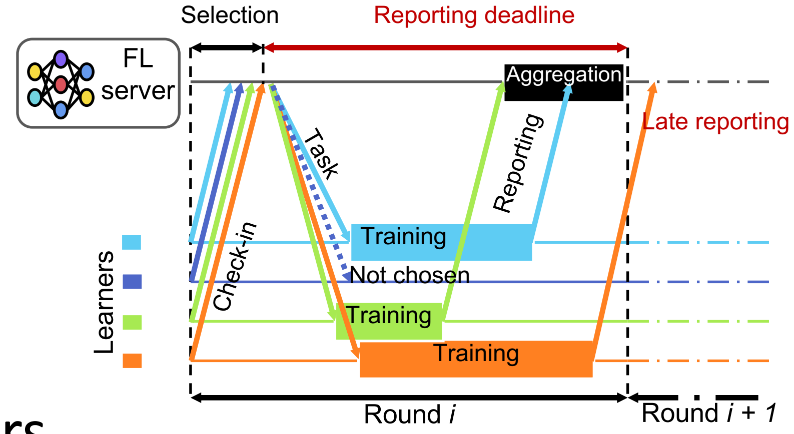
by **Karen Hao**   December 11, 2019

Medical Imaging



Ng D, Lan X, Yao MM, Chan WP, Feng M. Federated learning: a collaborative effort to achieve better medical imaging models for individual sites that have small labelled datasets. Quant Imaging Med Surg. 2021

Federated Learning Life-cycle



- FL server coordinates the learning stages

1. Online Learners check-in with the server
2. Participants Selection Stage:
   - The server selects a pre-set target number of learners
3. Updates Reporting Stage:
   - The selected participants perform the training task on local dataset
   - The server aggregates the updates to produce the global model

Heterogeneity is the grand challenge in face of Practical and Efficient FL!

*"Training in **heterogeneous** and potentially **massive networks** introduces novel challenges that require a fundamental departure from standard approaches for large-scale machine learning, distributed optimization, and privacy-preserving data analysis"*

In Federated Learning: Challenges, Methods, and Future Directions, T. Li, A. K. Sahu, A. Talwalkar and V. Smith., IEEE Signal Processing Magazine, 2020

Impact of Heterogeneity on Federated Learning

ACM EuroMLsys 2022

Work with Elton, Pantelis, Bilal, Marco @ KAUST

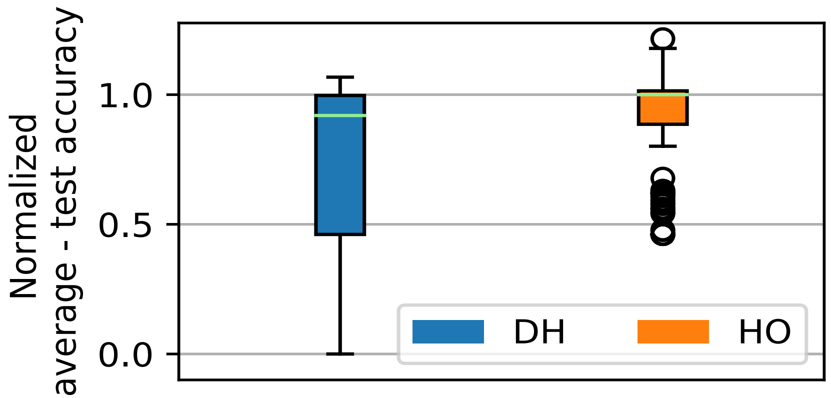
https://dl.acm.org/doi/10.1145/3517207.3526969

Heterogeneity in FL

- **Data** Heterogeneity:
  - Non-IID and skewed data distributions
  - **Many work focus on this** and mitigation methods have been proposed
- **Device** Heterogeneity:
  - Clients use different device models (heterogenous compute and network).
  - Time-to-accuracy rely on learners' training capabilities.
    - Clients with faster compute/network are likely to submit model updates before deadline.
- **Behavior** heterogeneity:
  - The process is driven by the participation/availability of the learners.
  - learners' availability vary over time due to imposed conditions on participation
    - Available if idle, connected to WiFi, and/or connected to power source

Impact of Heterogeneity

- Can heterogeneity impede the learning process?
  - Data, Device, Behavioral, Participation, Configuration, etc.
- Empirical study of the impact of heterogeneity on FL [1]
  - 1.5K experiments → varying models, datasets, FL & learning hyper-parameters, device proportions, aggregation algorithm, etc
  - We compare the following settings:
    - Homogenenous (HO): all devices are of the same configuration
    - Device Heterogenity (DH): Low-end, mid-range, high-end
- The impact is significant (**the model may not converge**)
  - Device heterogeneity degrades:
    - Model quality by up to 4.9X



More Extensive Analysis and Detailed Results in the paper

[1] Abdelmoniem et. al., Empirical analysis of federated learning in heterogeneous environments , ACM EuroMLSys'22

Mitigating The Impact of Heterogeneity

ACM EuroMLSys 2021

Work with Marco @ KAUST

https://dl.acm.org/doi/10.1145/3437984.3458839

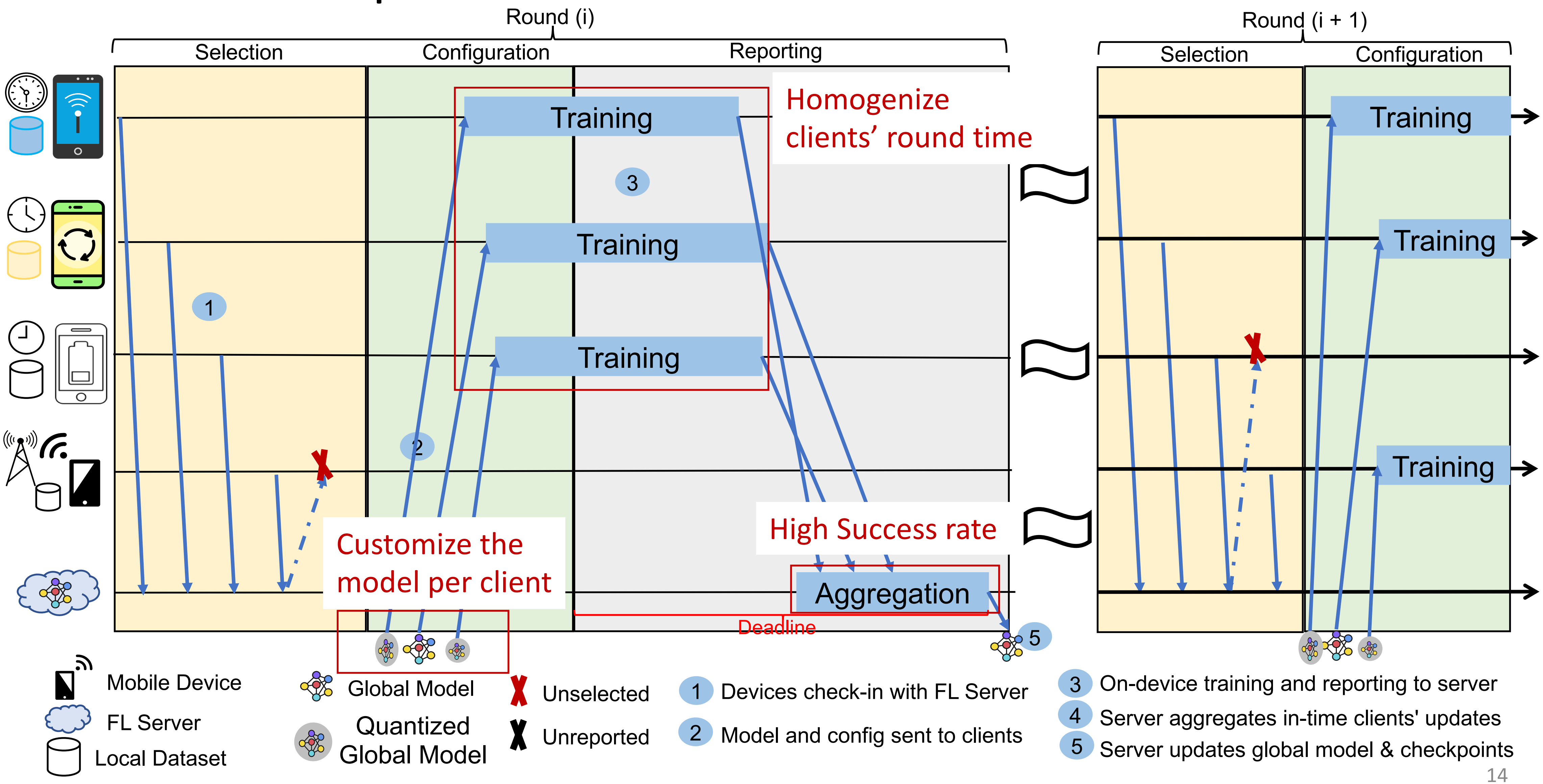
Full Version: https://arxiv.org/abs/2102.07500

AQFL: Adaptive Model Quantization

- Opportunity to mitigate device heterogeneity:
  - Devices vary significanlty in computational capability:
    - low-end (e.g., Nexus 6) to high-end devices (e.g., Huawei P50 Pro, Pixel 6)
    - Benchmarks of devices in the wild (e.g., https://ai-benchmark.com/ranking.html)/)
  - Hardware and software support for model quantization
    - IoT, Edge, and mobile devices (https://ai-benchmark.com/ranking_processors.html)
  - Model quantized training (Quantization-aware training):
    - Reduces upload/download time of the model
    - Reduces the computational needs to perform the training

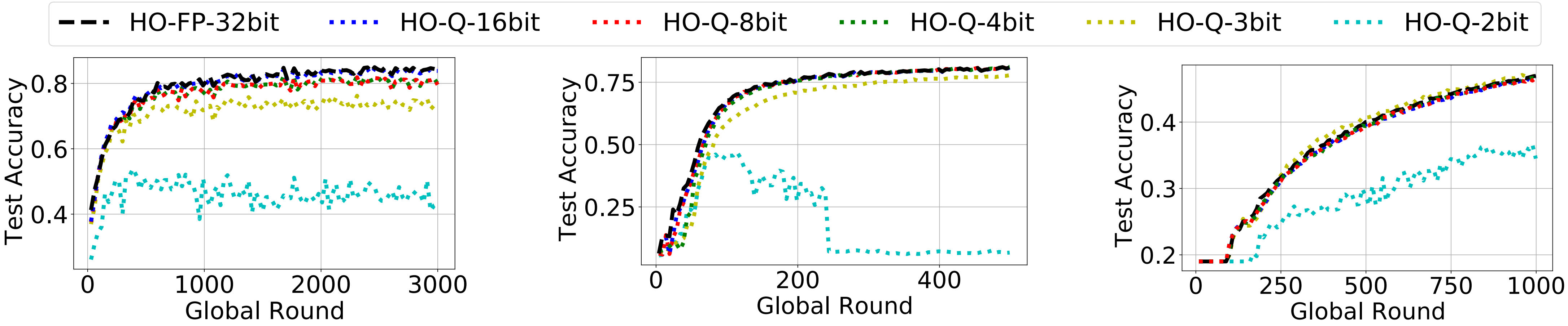
FL Server adaptively selects model quantization level (1-32bit) to homogenize round time of the heterogeneous devices

AQFL: Adaptive Model Quantization



Challenge: Quantization vs Quality

- Evaluate the quality degradation in homogeneous FL setting
  - Popular FL benchmarks and datasets - (CNN, RNN and LR) models.
  - Model quantization using Tensorflow's quantize library:
    - Quantize function - Inserts quantization operations in the computational graph
  - Acceptable quality for bit-widths >= 4 bits



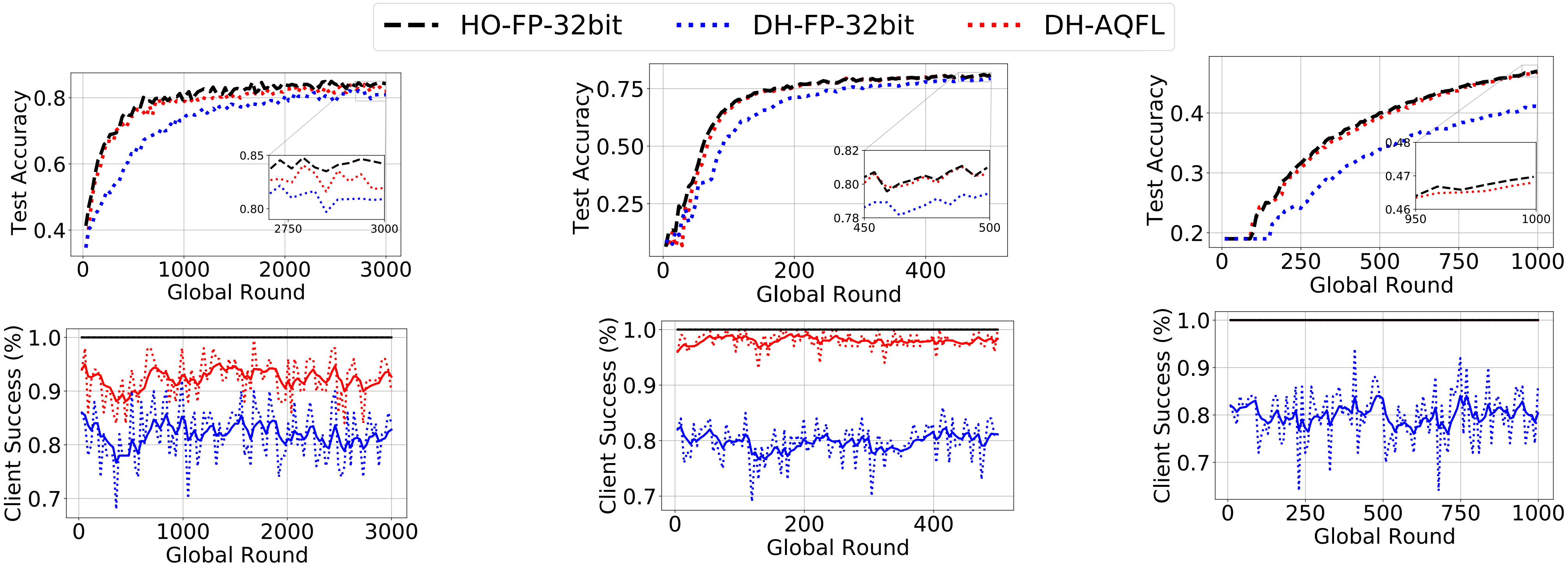
Logistic Regression - Synthetic

CNN - FEMNIST

RNN - Shakespeare

AQFL: Adaptive Model Quantization

- Server adaptively selects quantization level to homogenize round time
- Improves model quality and fairness:
  - Success rate of clients not missing the deadline is improved



Logistic Regression - Synthetic
CNN - FEMNIST
RNN - Shakespeare

REFL: Resource-Efficient Federated Learning

ACM EuroSys 2023

Work with Atal, Marco and Suhaib @ KAUST

https://arxiv.org/abs/2111.01108
https:// dl.acm.org/doi/10.1145/3552326.3567485
CODE: https://github.com/ahmedcs/REFL

Status Quo

- Many FL systems aim to improve the quality of FL trained models.
- Goal Reduce Time to Accuracy: Reduce Time ⬇ + Improve Quality ⬆

FedProx **(MLSys 2020), Yogi** **(ICLR 2021)**

- Improve the statistical efficiency of the learning process

Oort **(USENIX OSDI 2021)**

- Bias the client selection to Reduce the training time by exploiting the fast learners.

SAFA **(IEEE ToC 2021)**

- Invokes all learners for training and allows semi-synchronous model updates to boost the statistical efficiency

FLEET **(ACM Middleware 2021)**

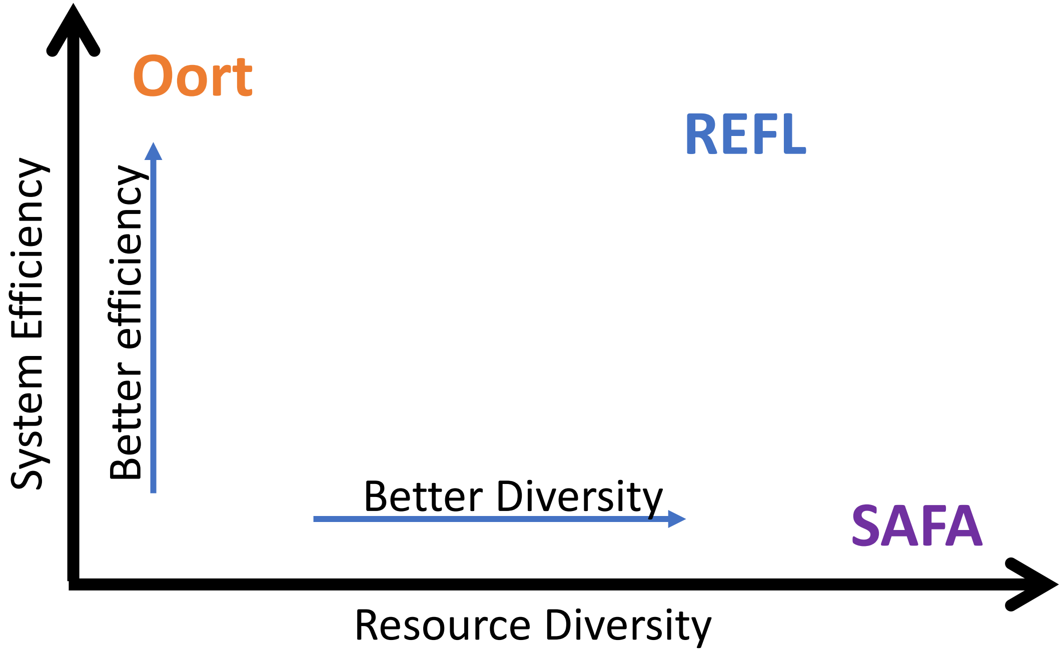
- Boost statistical efficiency via asynchronous updates with damping and boosting rules.

Motivation – Resource Efficiency

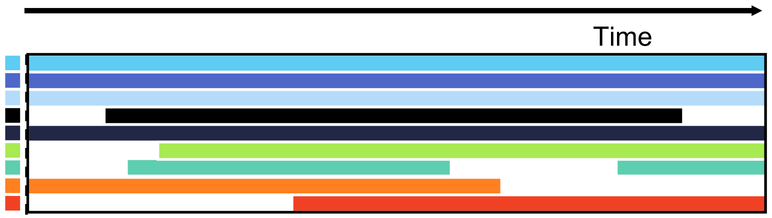
- We identify a Trade-off between System Efficiency vs Resource Diversity
  - Oort: favor fast learners over slow ones → high efficiency & low diversity
  - SAFA: select every learner → high diversity & low efficiency
  - Existing systems ignore resource consumption of the learners
- Our goal is to strike a balance between the two extremes

REFL: Resource Efficient FL System

- ❑ Increases the **diversity** by prioritizing selection of least available learners
- ❑ Aggregates stale updates to boost the statistical **efficiency**
- ❑ Leads to reduced **resource** consumption

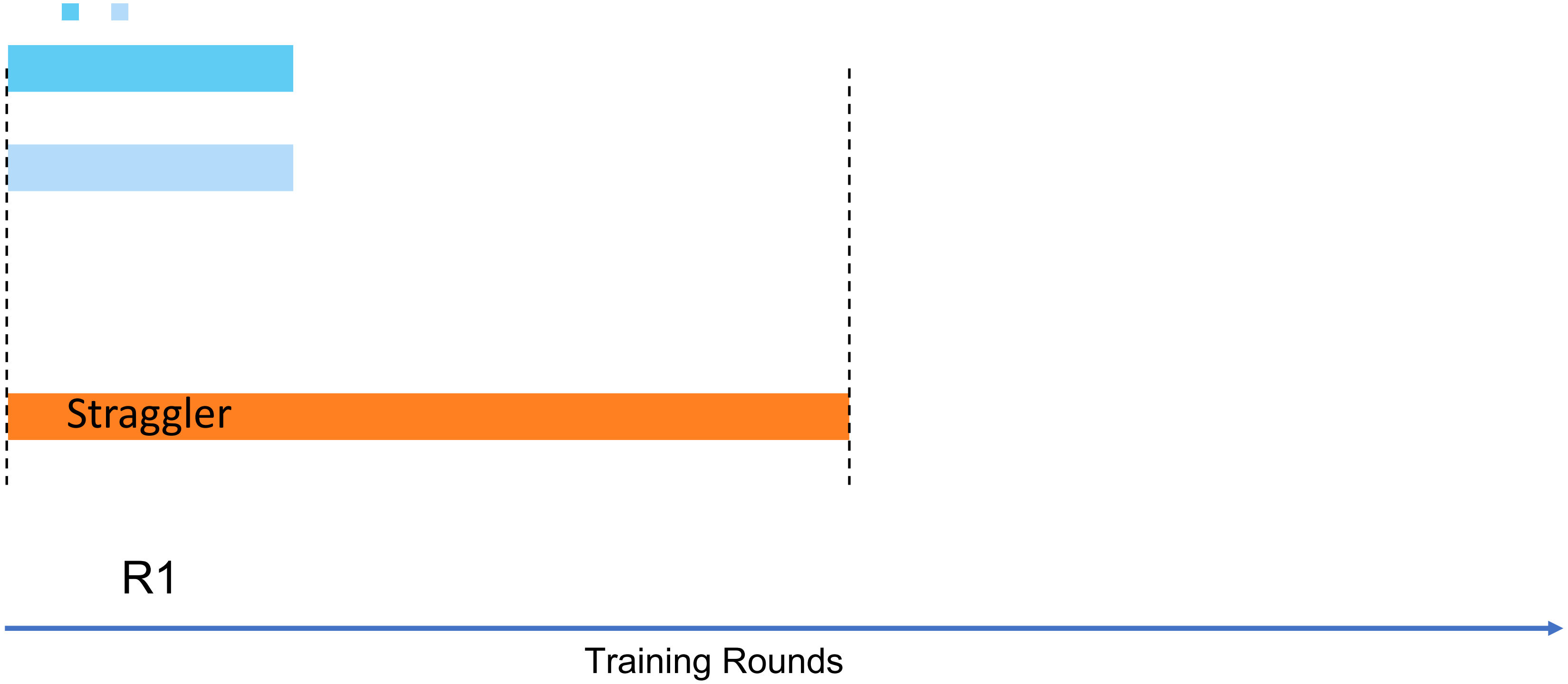


An Illustrative Example



(a) Learner availability

REFL selects among online clients using availability info and breaks ties by selecting at random



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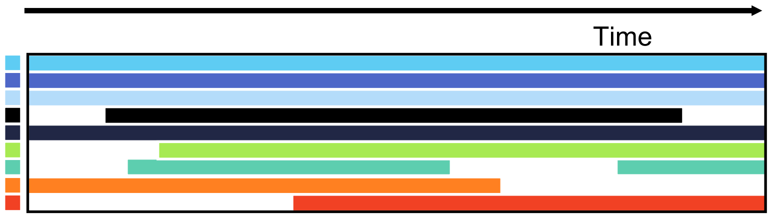

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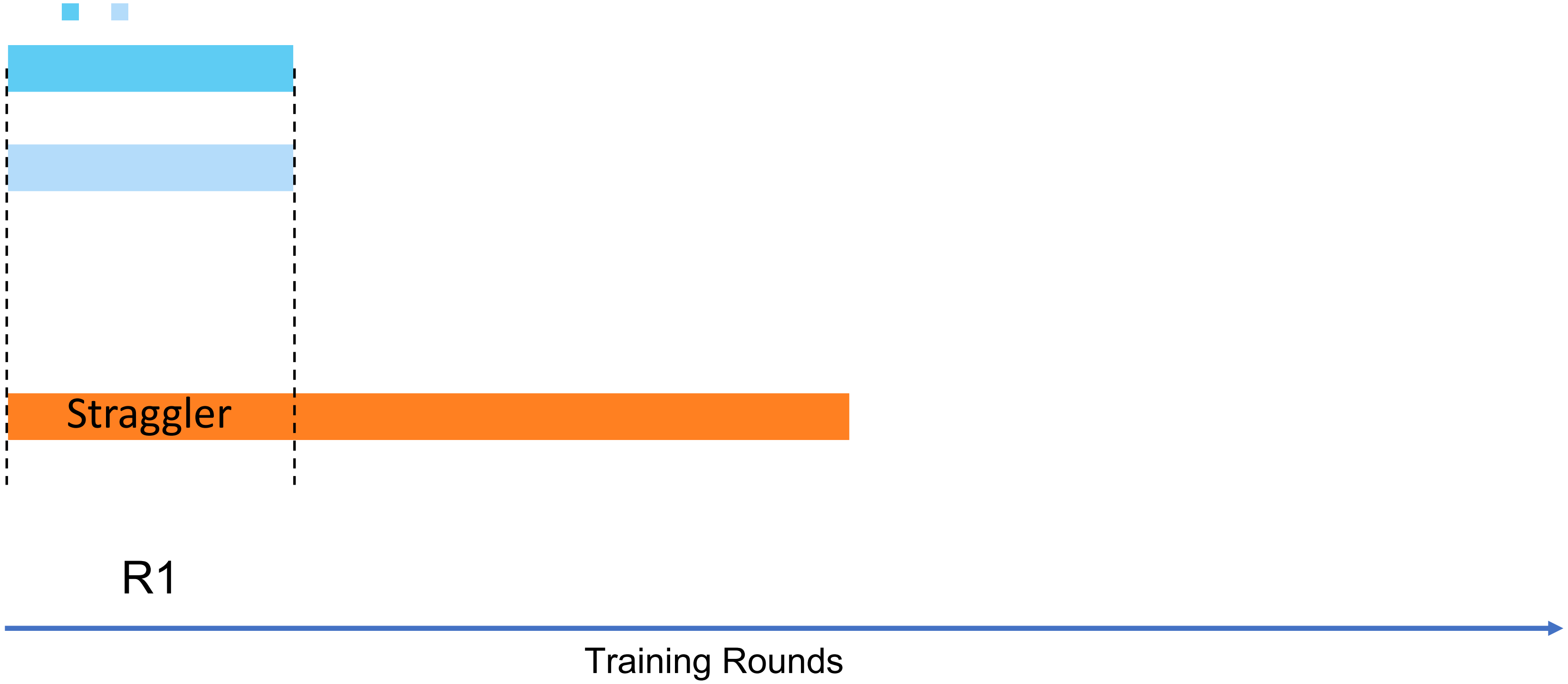

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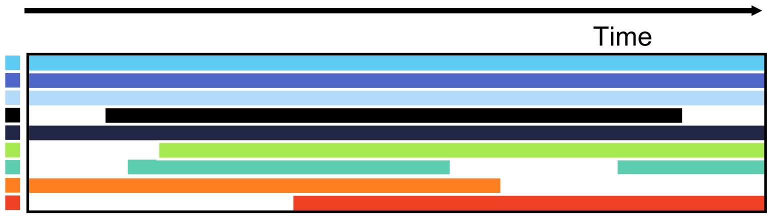
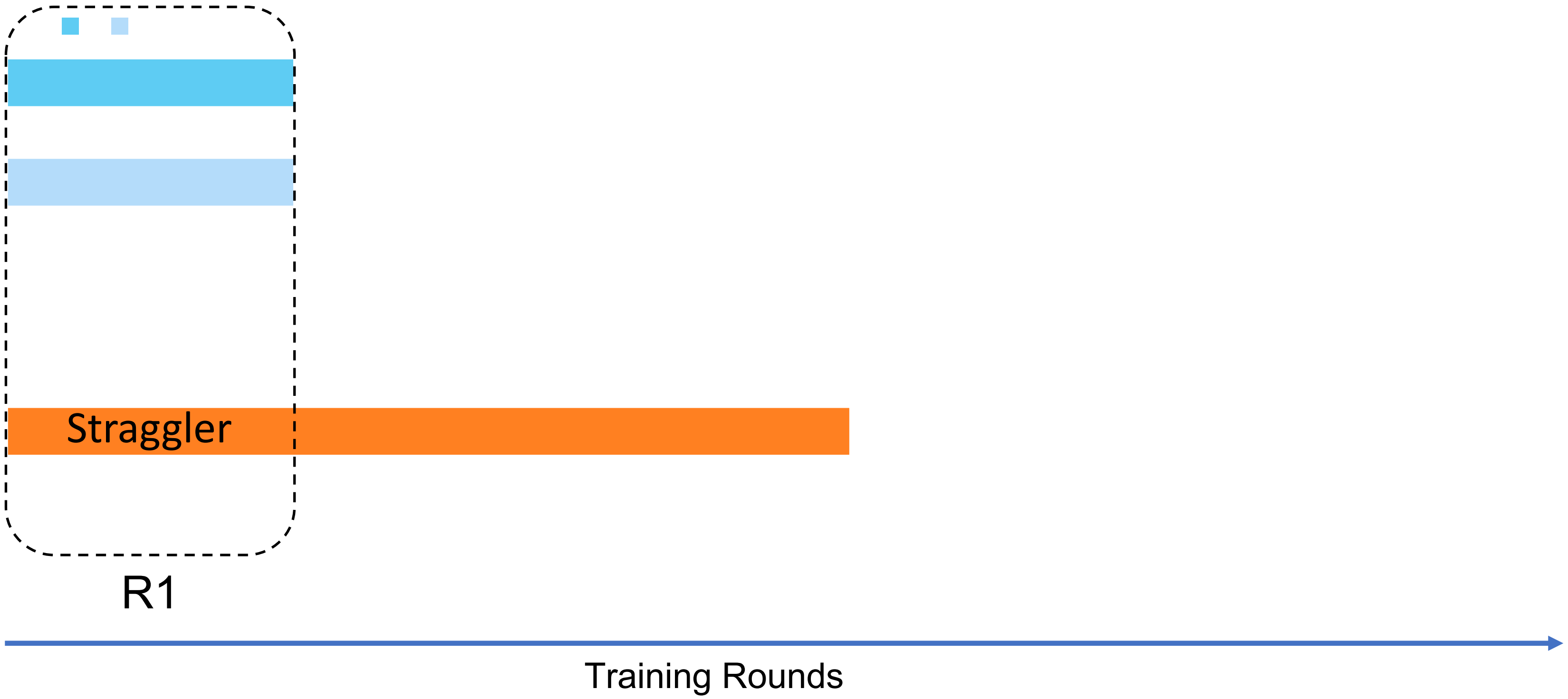
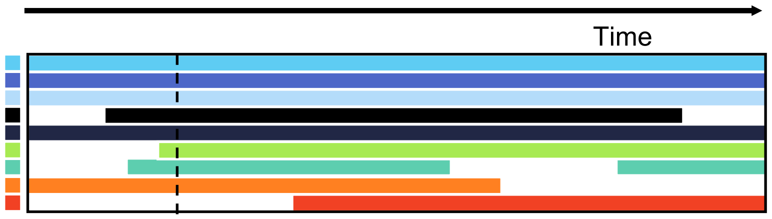


(a) Learner availability

REFL allows stragglers to continue their training, Oort would discard the slow learners

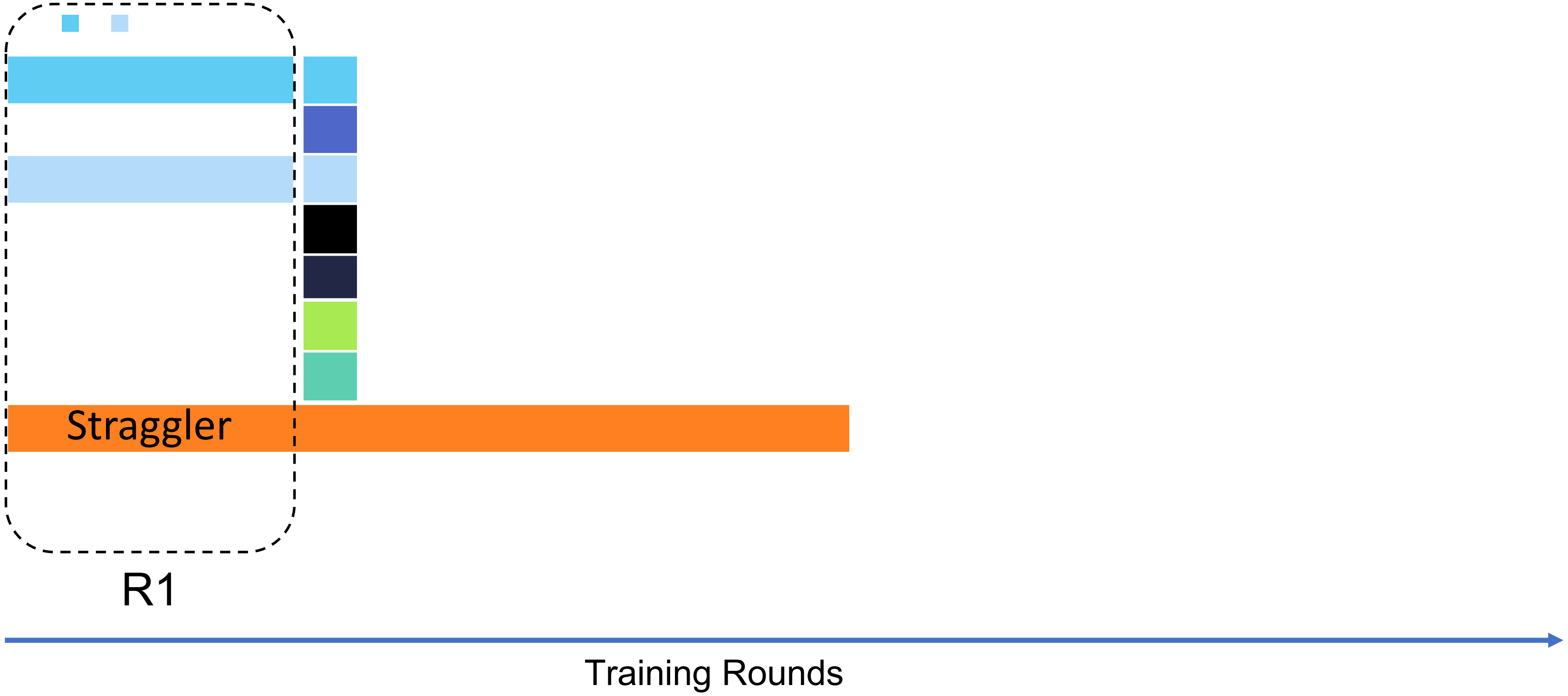


An Illustrative Example



(a) Learner availability

REFL selects the participants based on their future availability, SAFA would select all learners



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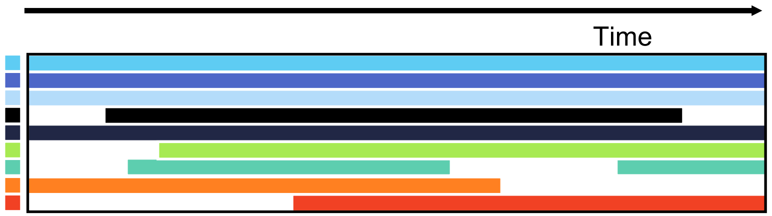

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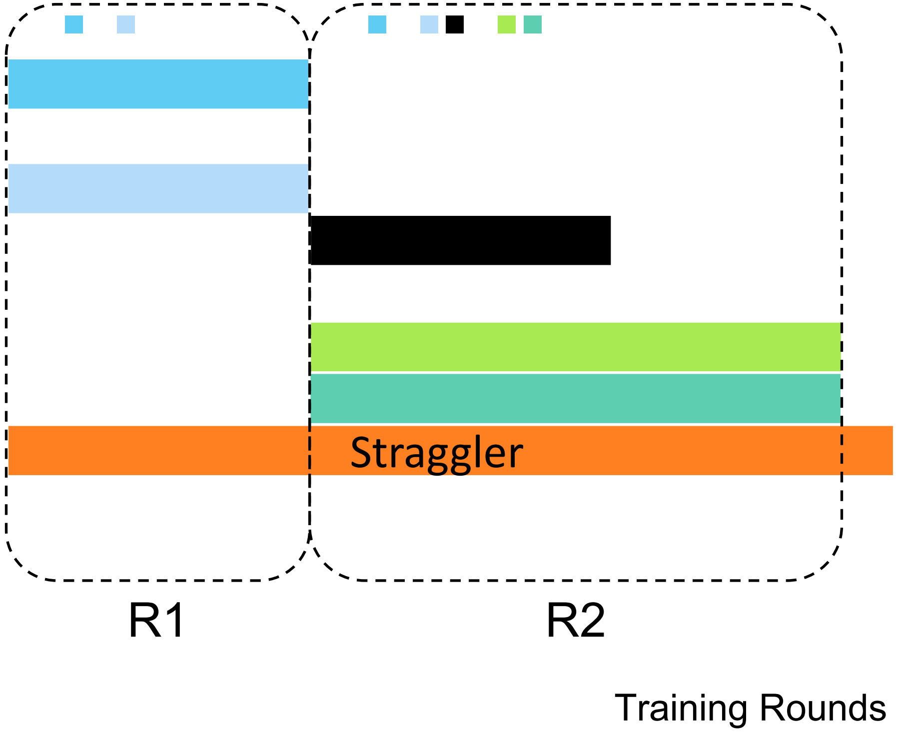

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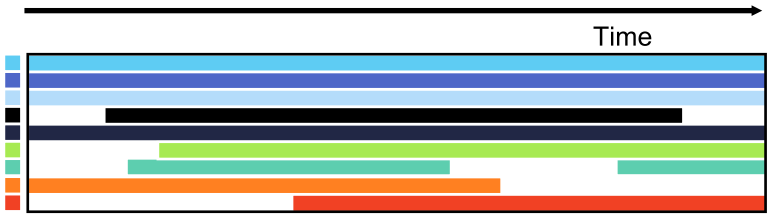
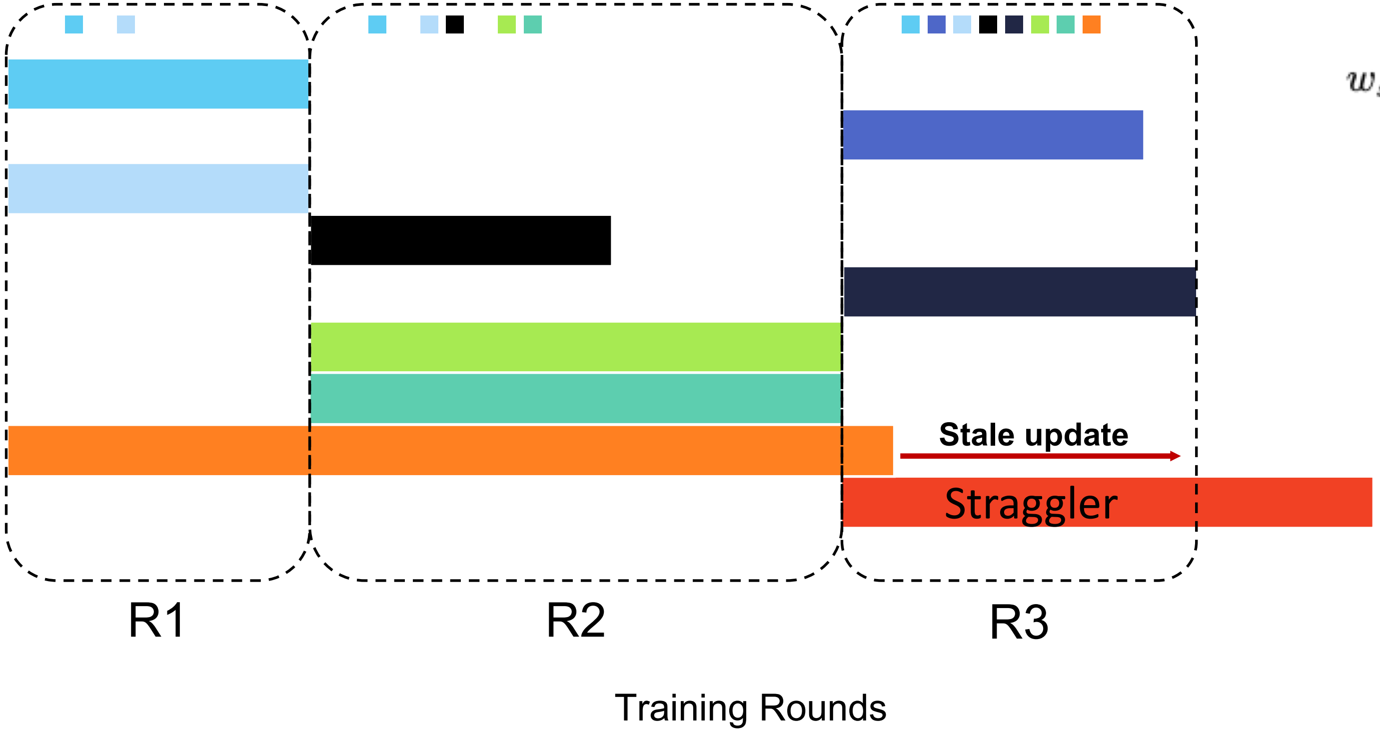


(a) Learner availability

REFL aggregates scaled version asynchronous updates to reduce resource wastage



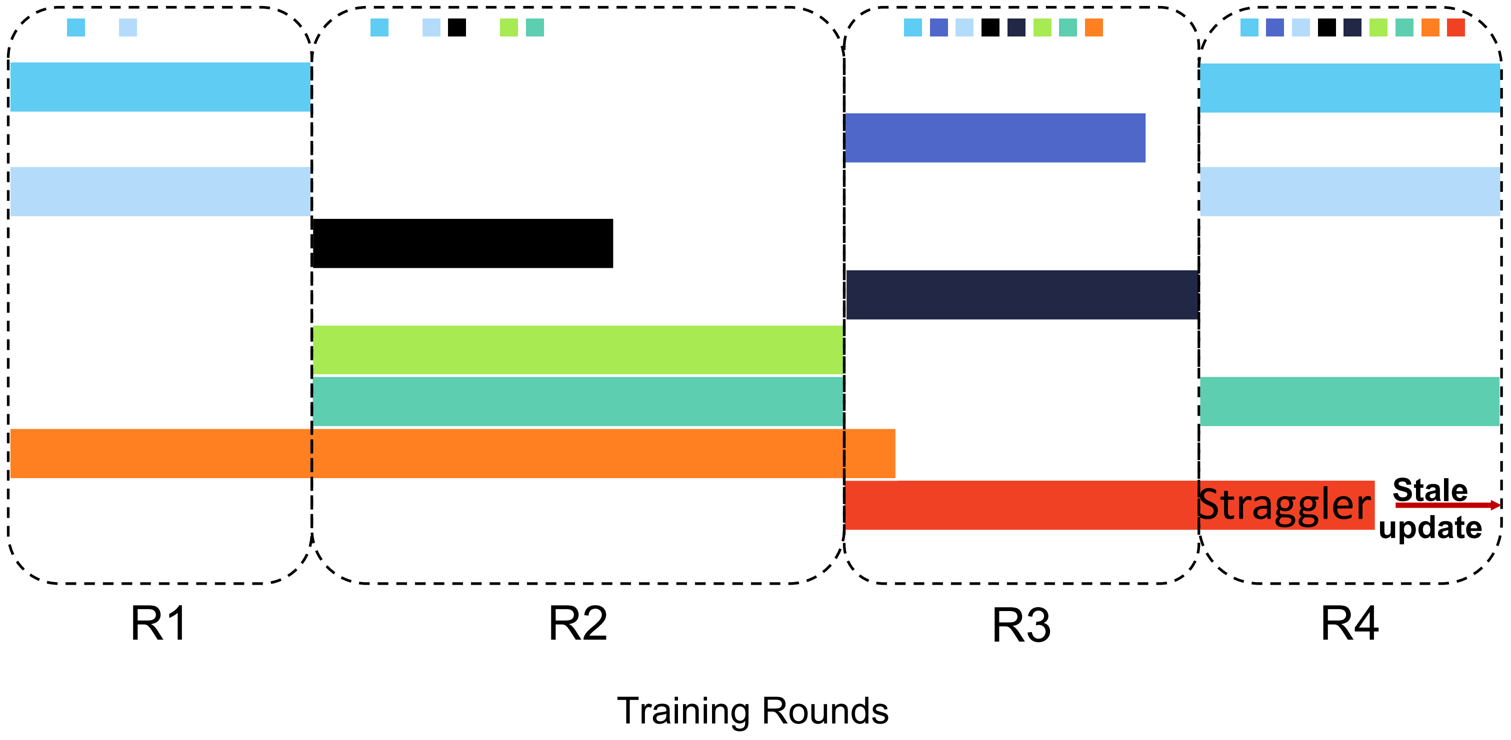
$$w_s = (1-\beta)\frac{1}{\tau_s+1} + \beta(1-e^{-\frac{\Lambda_s}{\Lambda_{max}}})$$

dampening boosting

An Illustrative Example

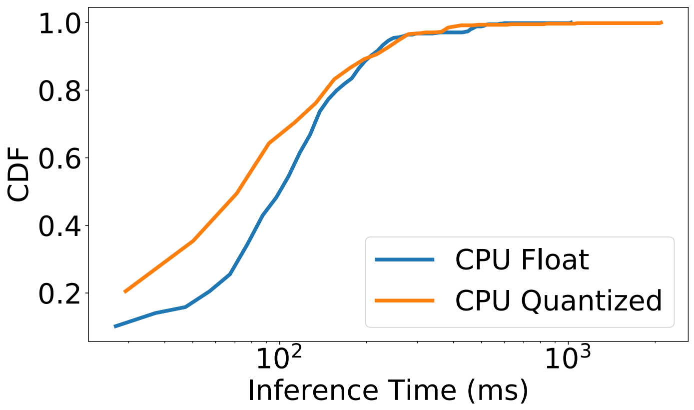
REFL

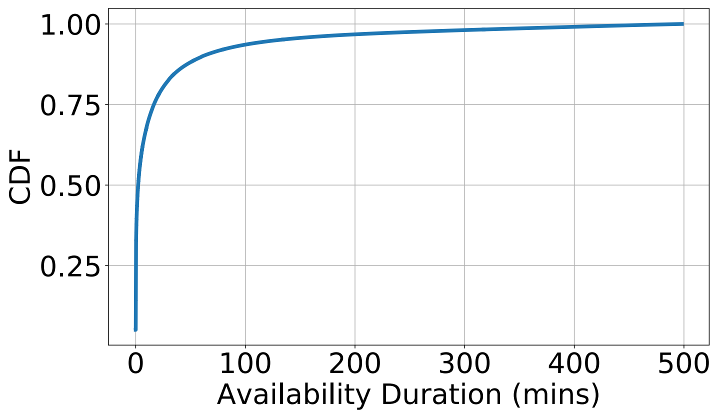
- **Higher rate of unique participants over Oort**
- **Lower resource wastage compared to SAFA**



Experimental Settings

- FL Benchmarks for various ML tasks
  - Speech Recognition, Computer Vision, Natural Language Processing
- Various data distributions: iid, FedScale [1], Label-limited (non-iid).
- Learners' **system**: AI benchmark (Compute) and MobiPerf (network)
- Learners' **availability**: User trace of 136K mobile clients





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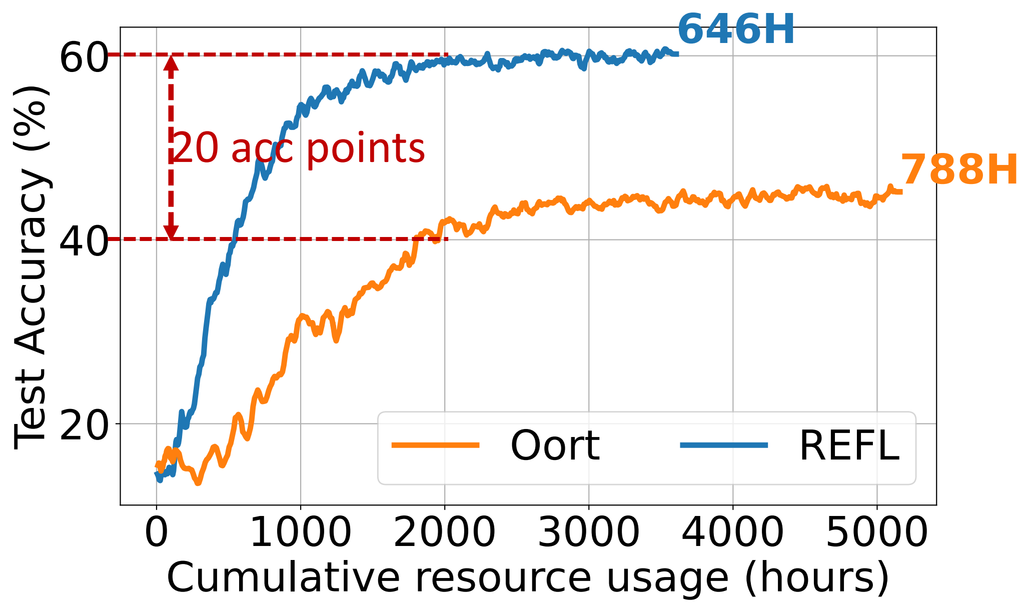
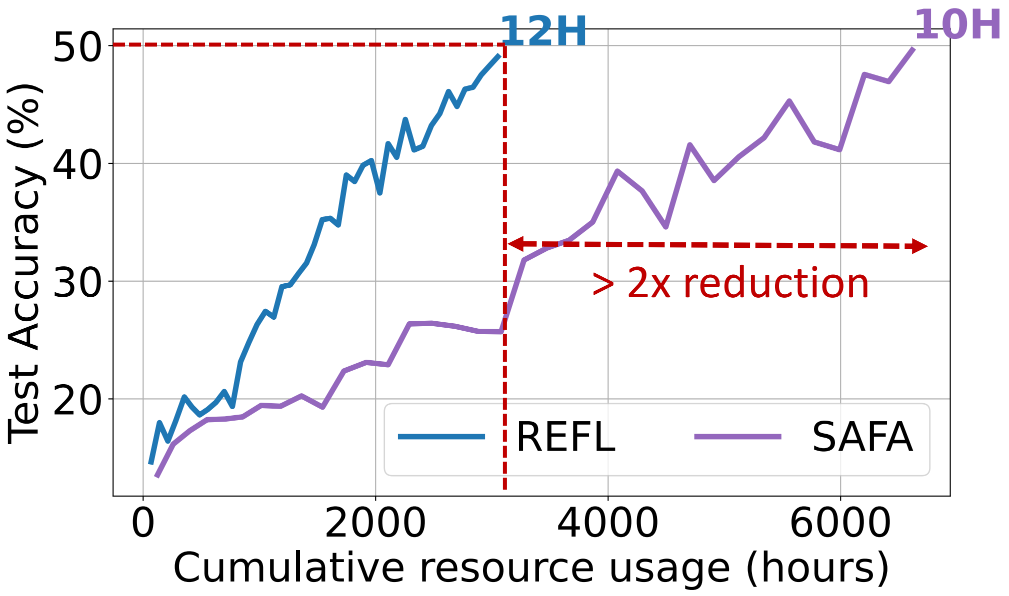
Evaluation of REFL

- REFL → best model quality with least amount of resources and time
  - **Availability prioritization** leads to better diversity.
  - **Aggregation of stale updates** leads to resource savings.

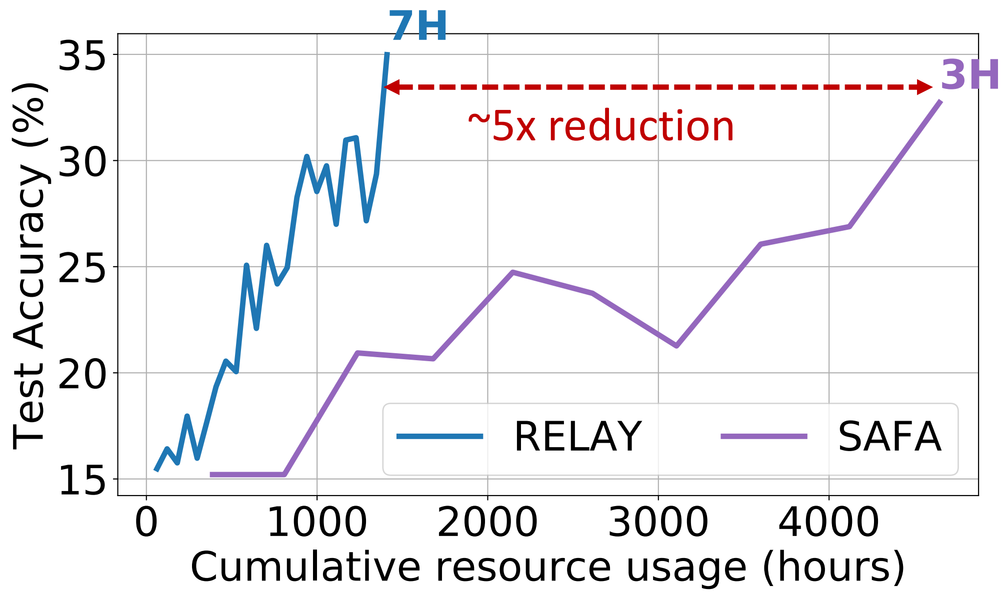




REFL is future-proof

Larger-scale FL scenario

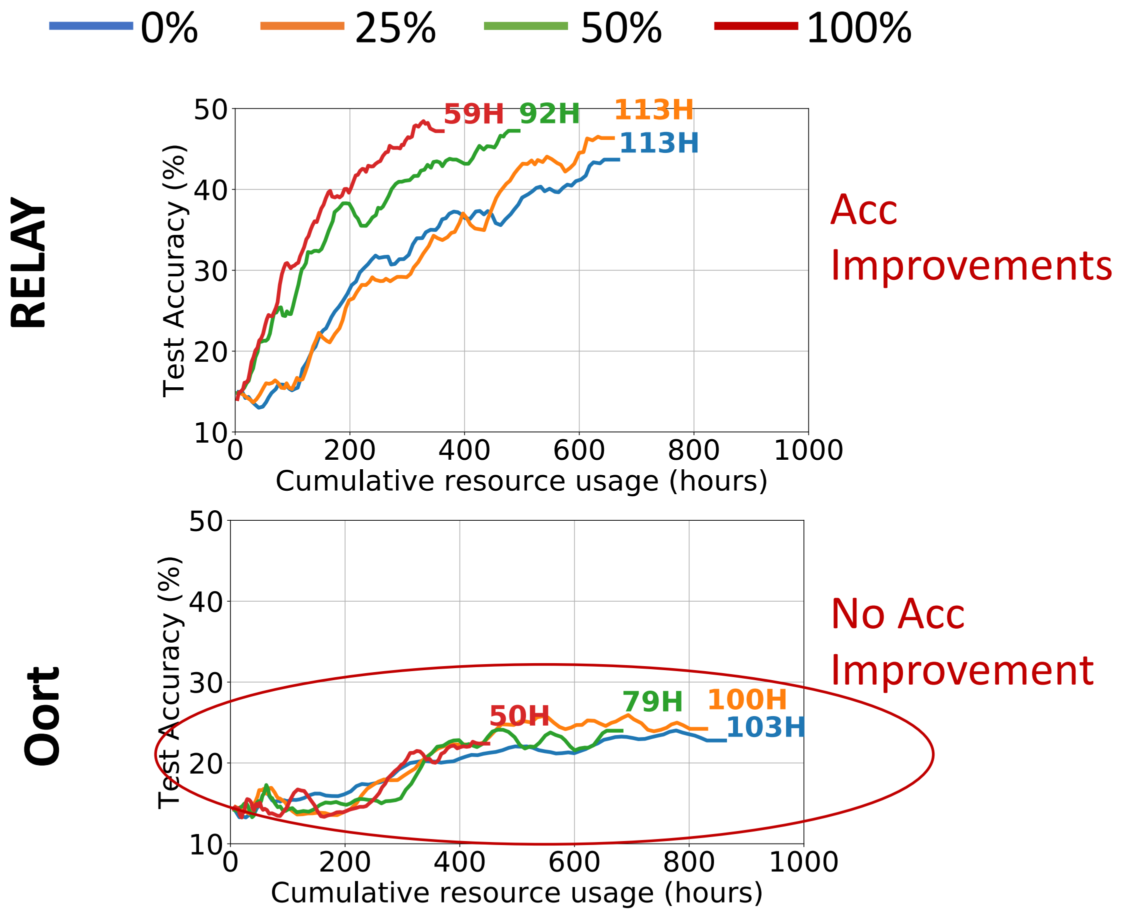
3x more learners



non-iid mapping

Hardware Advancements

Speed doubled for % of learners



non-iid mapping

Takeaways

- Heterogeneity is a major challenge for FL:
  - Model quality degradations are not acceptable, esp. with diverging cases
  - Device heterogeneity impact the quality the most.
- To tackle heterogeneity → adapt to available system HW/configs/dynamics
  - **AQFL** leverages support of on-device quantized training and customizes the models to increase success rate → gains in model quality
- Need to strike balance between resource usage and diversity
  - **REFL** innovates on **client selection** and **stale updates aggregation** to yield a resource efficient FL system → gains in quality, time and resource consumption!

Thanks

Q & A

To follow-up, please reach me at ahmed.sayed@qmul.ac.uk

If Interested in solving real-world problems!
Get in Touch!

For recent publication list see references [1-40]
https://scholar.google.com/citations?user=CzfuSJgAAAAJ&hl=en

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